AnalogAtlas | System Architecture

A reproducible science build feeds a small, static application that works offline.

LOCAL INPUTS

Earth terrain

GEBCO_2026 15-arcsec grid, or a documented official replacement.

Moon terrain

LOLA polar DEM candidate; verify labels, posting and target coverage.

Mars terrain

MOLA MEGDR 128 pixels/degree for the Jezero reference.

Frozen configuration

Targets, bounds, grid origin, gates, methods and source versions.

Site registry

Cited analog examples and contrast controls, kept out of calibration.

PYTHON SCIENCE BUILD

1. Normalize inputs

Adapters provide elevation, valid-data mask, units and source lineage.

2. Project

Local metric Lambert azimuthal equal-area; body-specific planetary radii.

3. Common grid

Area-weighted elevation resampling to 500 m square cells.

4. Terrain

Horn 3x3 slope; 1 km derivative baseline; 1.5 km stencil footprint.

5. Support features

Four summaries on fixed 16 x 16 km (256 km2) supports.

6. Compare and validate

Discovery-only IQR tolerances; deterministic score; held-out validation after freeze.

PRODUCT OUTPUT

Science tables

Parquet support features, scores and exclusion reasons.

Compact app data

Local JSON / GeoJSON for targets, top candidates, registry and validation.

Run provenance

Source versions, config checksum, methods, caveats and run ID.

Static web app

React + Vite + MapLibre with local Natural Earth vectors.

Offline experience

Target selector, ranked map/list, candidate detail and datasheet.

No backend, hosted tiles, cloud processing, paid service or LLM is required at runtime.

Science Engine | Scale, Features and Score

All bodies use the same declared support geometry and interpretable terrain summaries.

PROCESSING SCALE

DEM + mask
Source elevations, nodata and lineage

→

Metric LAEA
Local equal-area projection

→

500 m grid
Area-weighted resampling

→

Horn slope
3x3; 1 km baseline

→

16 km support
32 x 32 cells = 256 km2

FOUR FEATURES - TWO EQUALLY WEIGHTED FAMILIES

Slope distribution | family weight 0.50
Median slope: typical inclination (degrees). Slope P90: upper-tail steepness (degrees). Each feature weight: 0.25 overall.

Terrain variability | family weight 0.50
Relief P95-P5 (metres) + population slope SD (degrees). Each feature weight: 0.25 overall.

Target vector
One fixed 16 x 16 km reference support per body: Connecting Ridge reference and Jezero landing neighborhood.

Tolerance, frozen before validation
For each feature: delta = 0.25 x mean of the three window-specific Earth IQRs. Same alpha for all features.

No target heterogeneity estimate
One support gives n=1. Spatial heterogeneity is not estimated or treated as zero; it is omitted.

Per-feature agreement
a = 1 - min(1, |candidate value - target value| / delta). Clamp each feature agreement to [0, 1].

Terrain Morphology Agreement
Score = 100 x sum(feature weight x agreement). A 0-100 index; not probability, safety, or mission suitability.

Eligibility and missing data
Require >=90% valid cells and all four features. Otherwise exclude with a reason; never renormalize around missing features.

IQR and alpha are normalization heuristics. Correlation is audited before validation; no post-validation tuning or PCA scoring.

Product | Validation and Team Build

What users inspect, what stays held out, and how the four-person team assembles the demo.

OFFLINE APPLICATION

Select reference

Moon or Mars; show the fixed support and source.

Explore candidates

Ranked map and list across three bounded Earth discovery windows.

Inspect datasheet

Four feature values, units, score contributions, coverage and lineage.

Assess evidence

Evidence Quality is separate: coverage, source lineage and sensitivity.

Read limitations

Scale, run provenance, sources, caveats and explicit non-claims.

DISCOVERY AND VALIDATION ARE SEPARATE

Discovery windows

East African Rift / Afar; Namib; Ladakh / western Himalaya. Used only to estimate Earth IQRs and rank candidates.

freeze first

Held-out site examples

Atacama / Yungay; Haughton Crater; Death Valley. Scored only after scientific freeze; site-target relevance is reviewed.

Contrast controls + report

Preselected controls are not true negatives. Show descriptive ranks, failures and limits; no ROC-AUC or predictive claim.

TEAM OF FOUR

Science / data

Sources, licenses, windows, targets and site labels.

Geospatial

Adapters, projection, common grid, masks and feature build.

Scoring / validation

Freeze, tolerance, deterministic score, controls and report.

Frontend / demo

Offline map, ranking, detail, datasheet and presentation.

48-hour sequence

Rules + schemas > vertical slice > all windows/targets > freeze > rank > validate > offline QA > demo